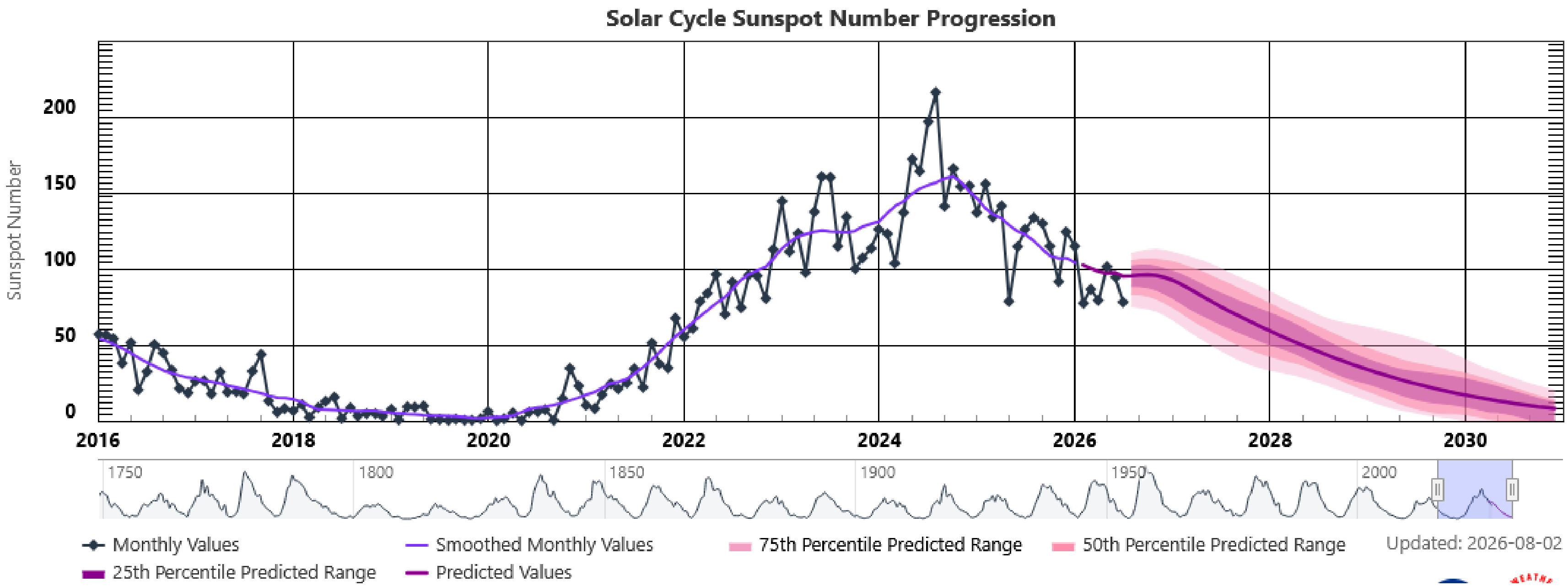
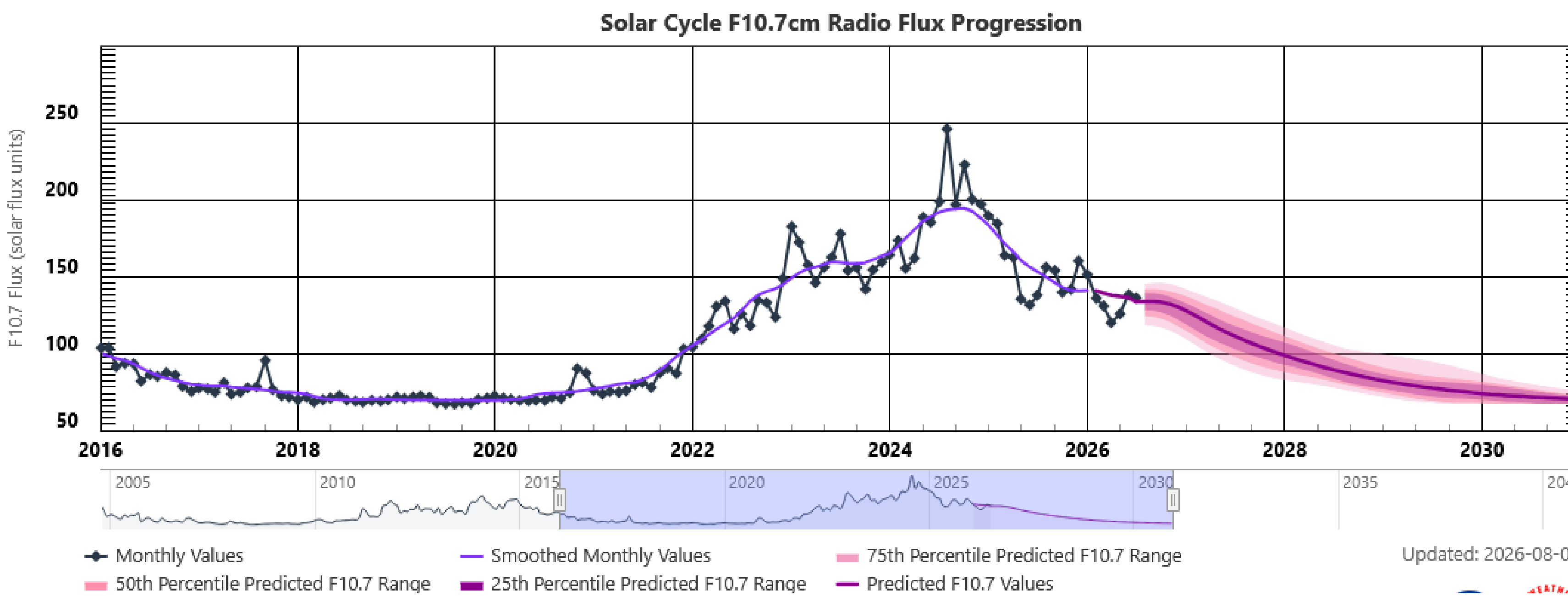
SOLAR WEATHER
4 AUG 2026

Lewis Thompson
W5IFQ

Taken by
Jessica Fridrich
on July 4, 2026
@ Binghamton,
NY

Taken by Thomas McCarty on
October 30, 2022 @ Fairbanks
Alaska


Solar Cycle Sunspot Number Progression
Sunspot Number
0
50
100
150
200
2016
2018
2020
2022
2024
2026
2028
2030
1750
1800
1850
1900
1950
2000
Monthly Values
Smoothed Monthly Values
75th Percentile Predicted Range
50th Percentile Predicted Range
25th Percentile Predicted Range
Predicted Values
Updated: 2026-08-02


Solar Cycle F10.7cm Radio Flux Progression
F10.7 Flux (solar flux units)
50
100
150
200
250
2016
2018
2020
2022
2024
2026
2028
2030
2005
2010
2015
2020
2025
2030
2035
2040
Monthly Values
Smoothed Monthly Values
75th Percentile Predicted F10.7 Range
50th Percentile Predicted F10.7 Range
25th Percentile Predicted F10.7 Range
Predicted F10.7 Values
Updated: 2026-08-02

SolarHam

Indices: (8/4 @ 00:35 UTC) SFI 126 ▼ 1 SSN 104

Geomagnetic Forecast

Aug 4	Aug 5	Aug 6
4 (G0)	2 (G0)	2 (G0)

Max Kp

Aug 4		Aug 5		Aug 6	
M-Lat	05%	M-Lat	05%	M-Lat	05%
H-Lat	25%	H-Lat	20%	H-Lat	15%

Probabilities

Solar Flare Probabilities

C-Flare: **90%** M-Flare: **20%**

X-Flare: **01%** Proton: **01%**

Probability Details

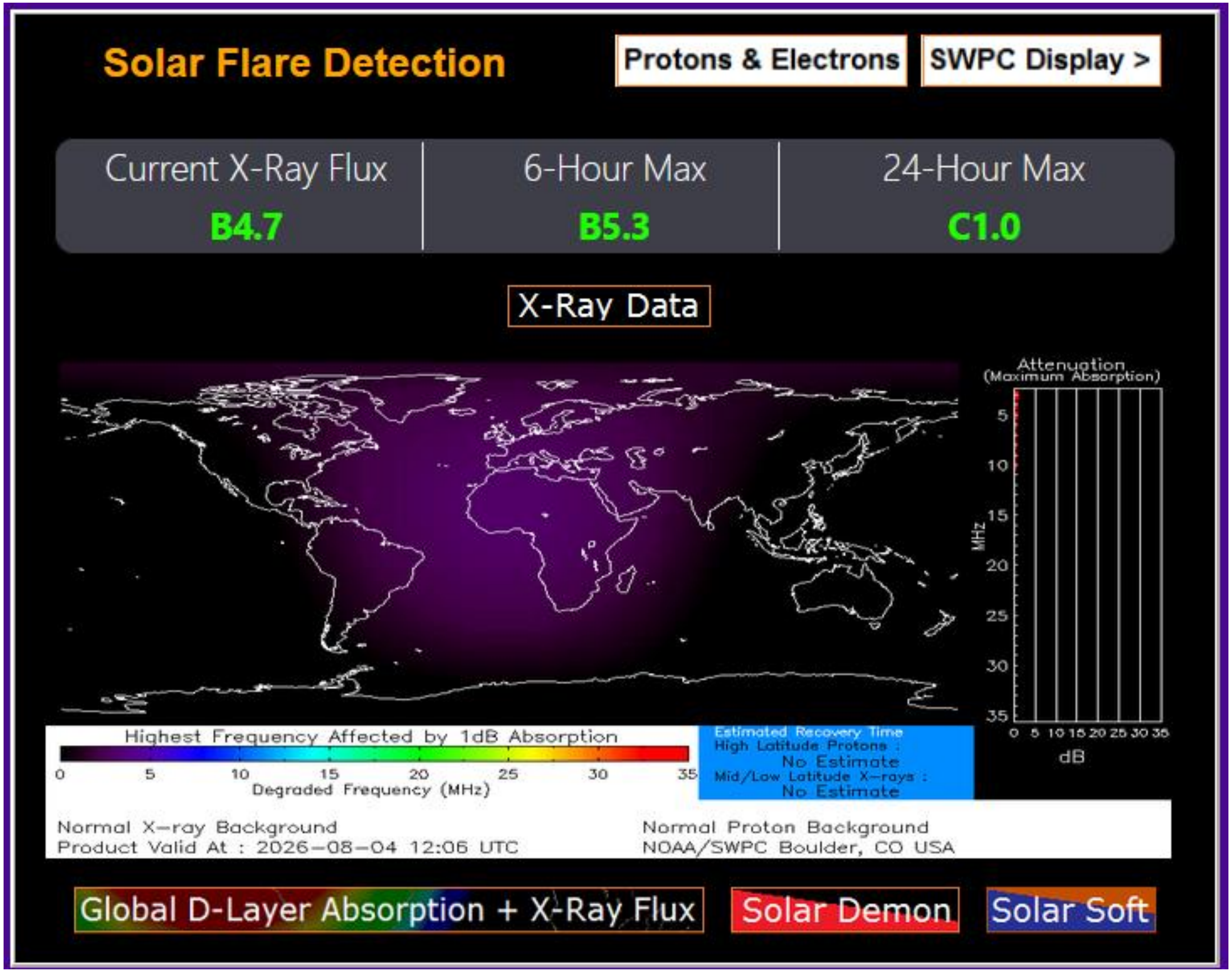
Updated @ 19:20 UTC (August 2)

Visible Sunspot Regions

AR 4502	B	S06W10	▲
AR 4501	B	S07W53	—
AR 4500	B	N18W12	▼
AR 4499	B	N04W51	—
AR 4498	BD	N14W17	▲
AR 4496	A	N04W77	—

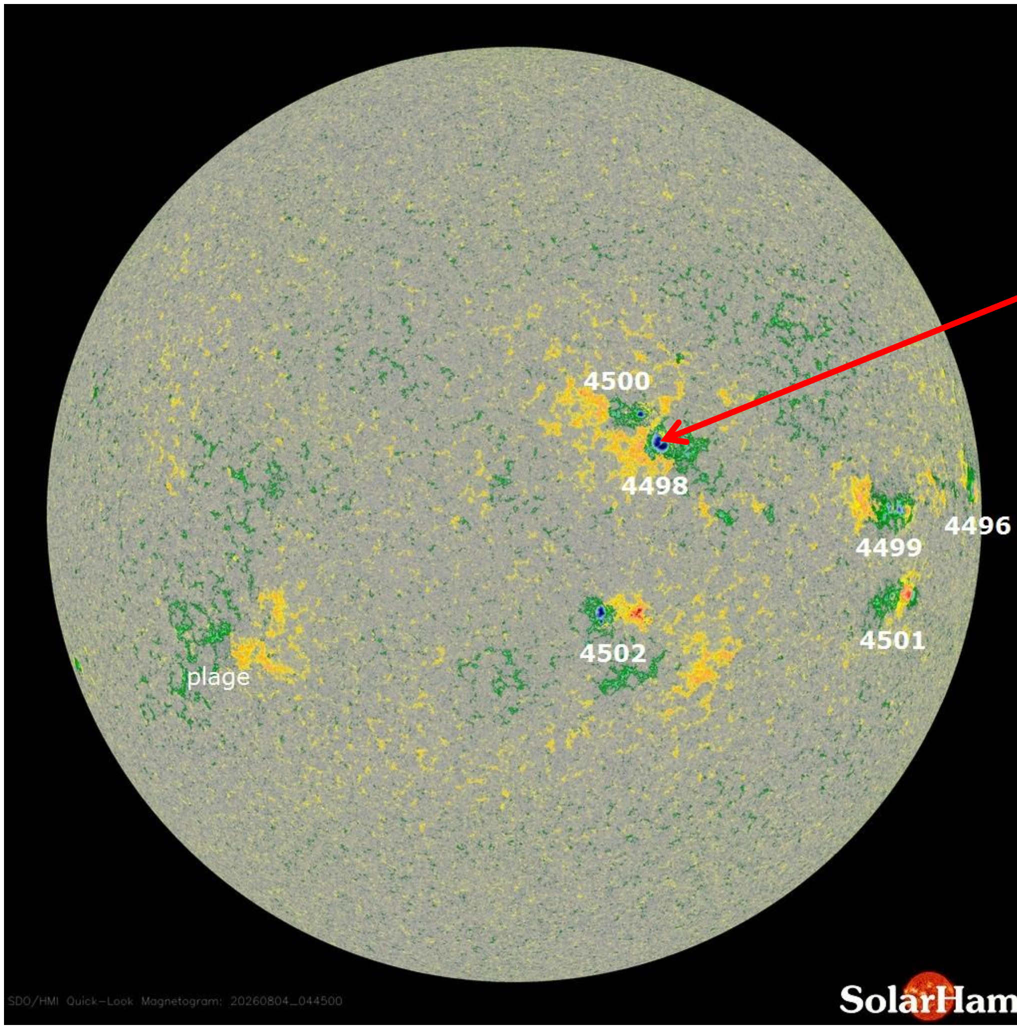
Updated @ 00:35 UTC (August 4)

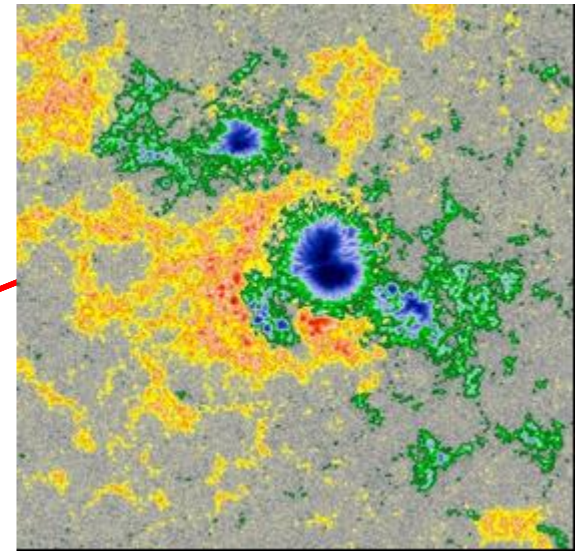
SolarHam


Solar Flare Detection
Protons & Electrons
SWPC Display >
Current X-Ray Flux
B4.7
6-Hour Max
B5.3
24-Hour Max
C1.0
X-Ray Data
Attenuation
(Maximum Absorption)
MHz
dB
Highest Frequency Affected by 1dB Absorption
Degraded Frequency (MHz)
Estimated Recovery Time
High Latitude Protons :
No Estimate
Mid/Low Latitude X-rays :
No Estimate
Normal X-ray Background
Product Valid At : 2026-08-04 12:06 UTC
Normal Proton Background
NOAA/SWPC Boulder, CO USA
Global D-Layer Absorption + X-Ray Flux
Solar Demon
Solar Soft

Sun Spots

Magnetogram Image (Updated August 4, 2026)

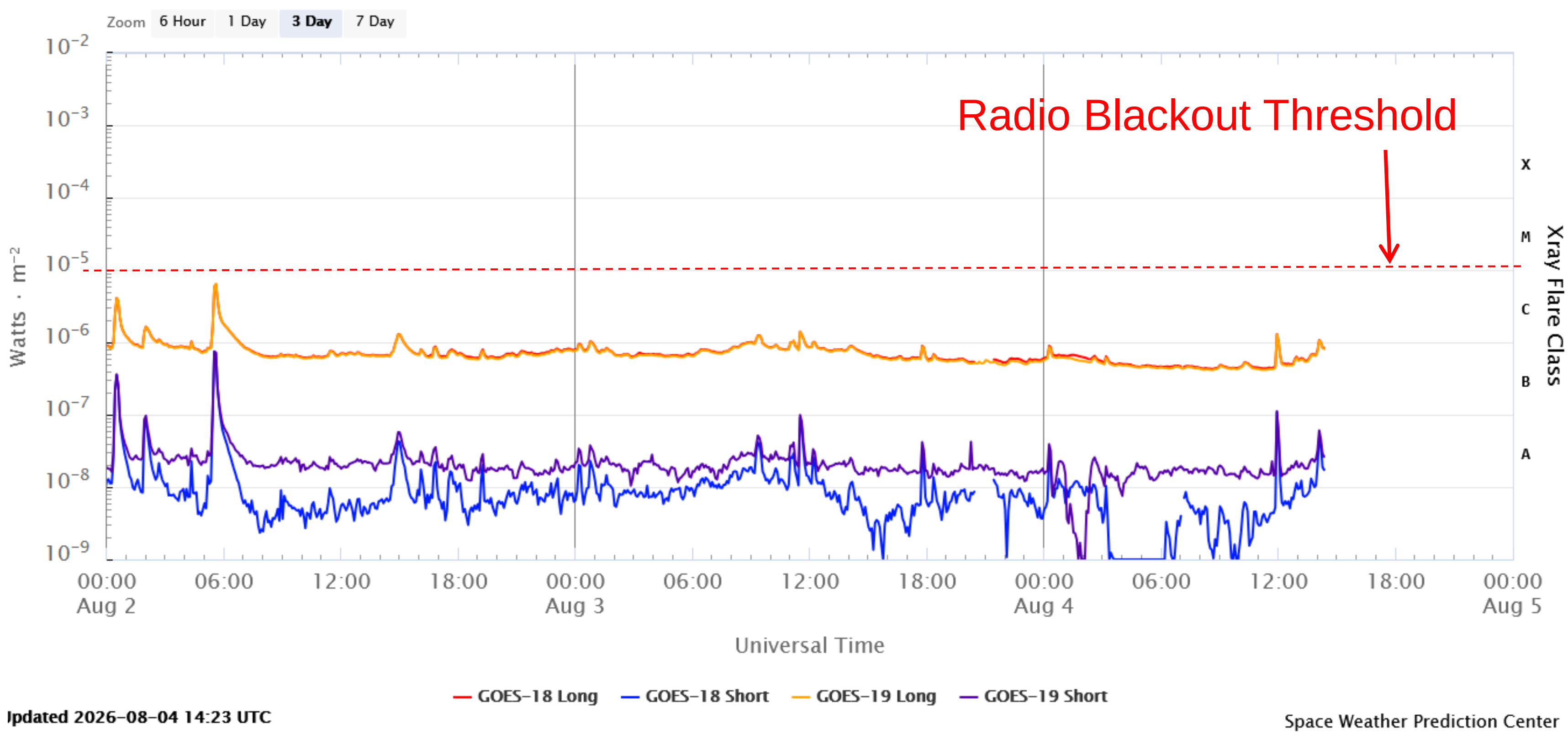




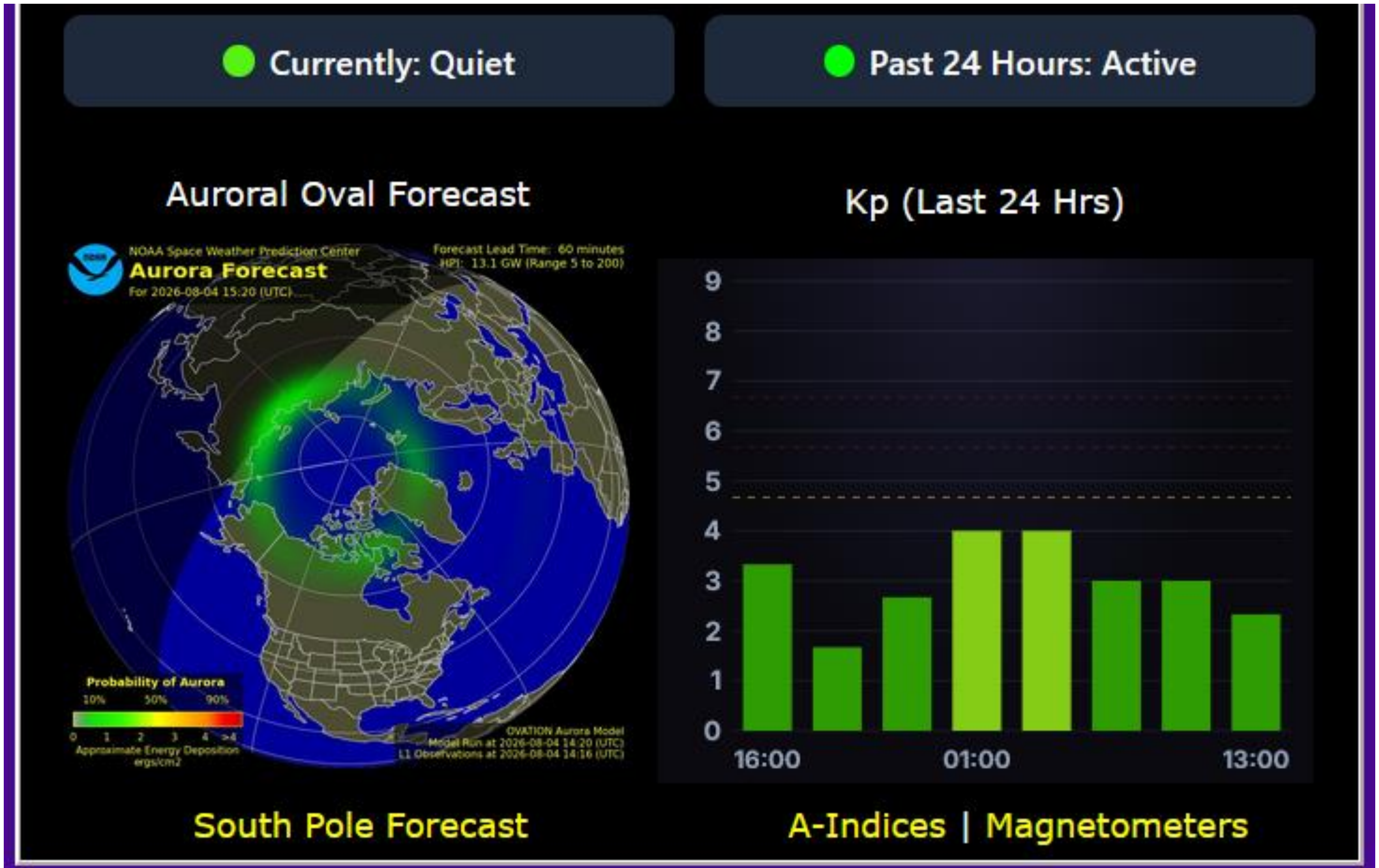
Beta Delta

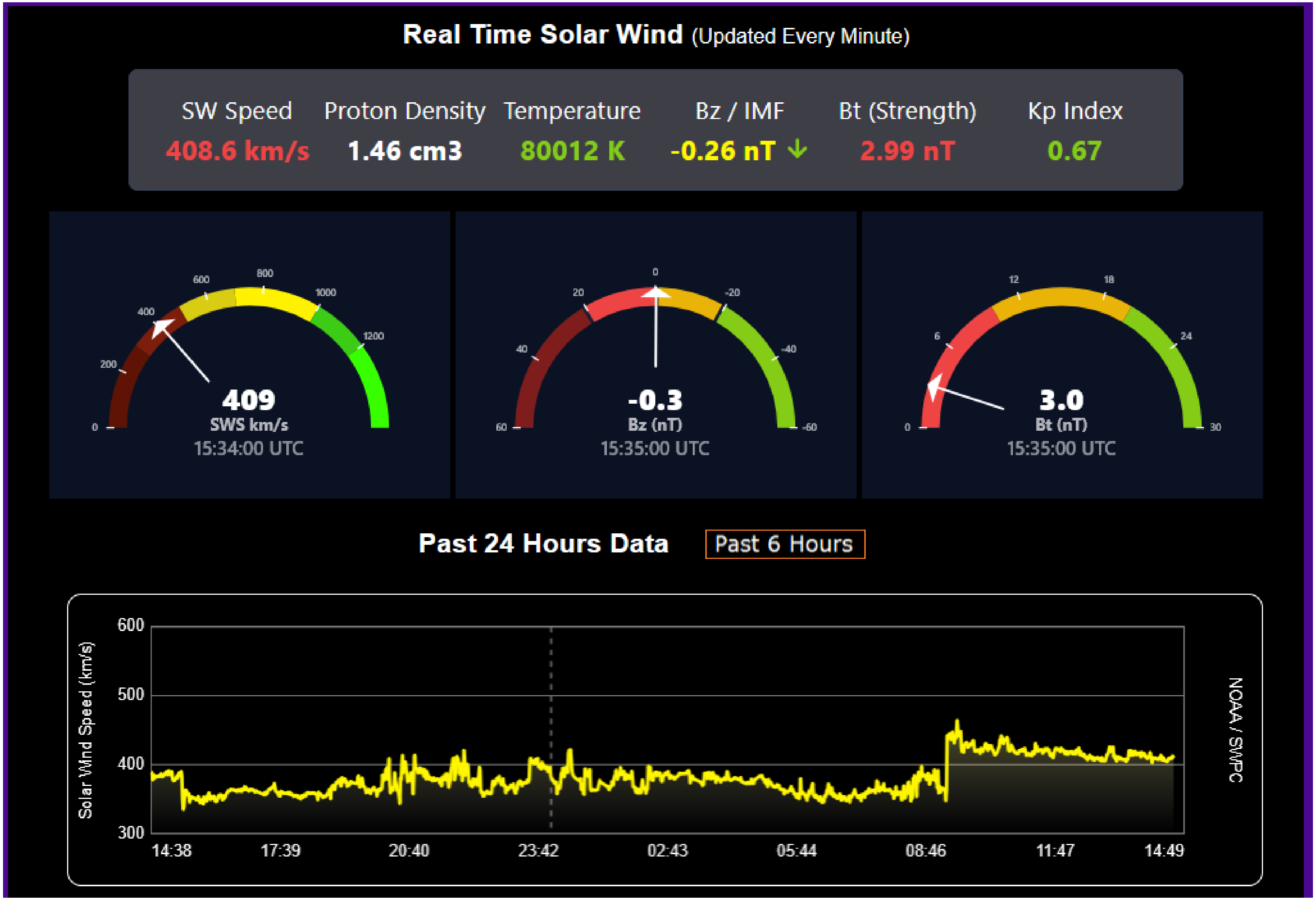
Location	Spot Count	Sunspot Area	
N14W17	16	180	
Flare Threat	C: 55%	M: 10%	X: 01%

Solar X-Ray Flux: 2 – 4 AUG 2026



Earth's Geomagnetic Activity
4 AUG 2026

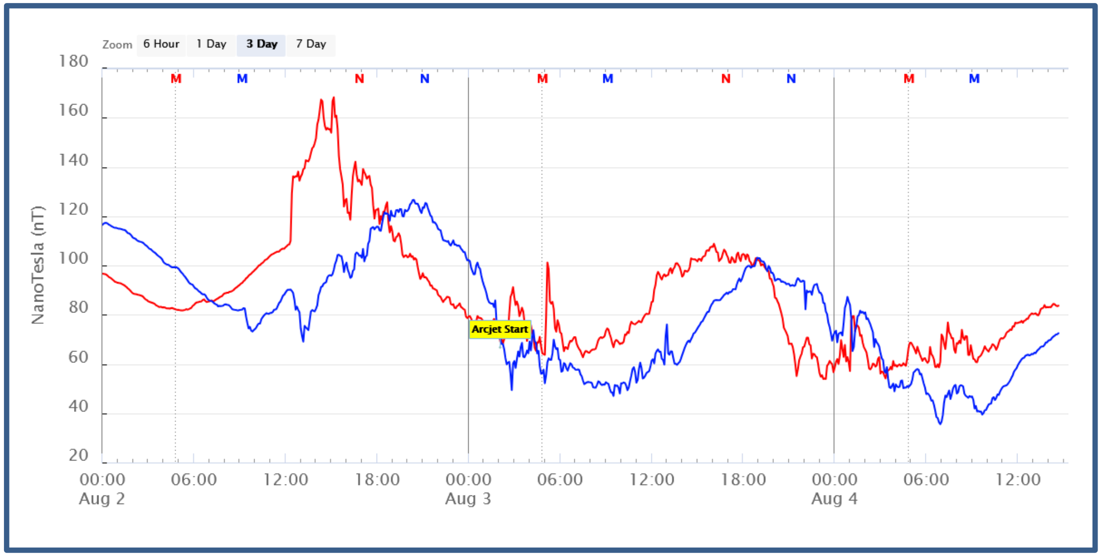



Real Time Solar Wind (Updated Every Minute)
SW Speed
408.6 km/s
Proton Density
1.46 cm3
Temperature
80012 K
Bz / IMF
-0.26 nT
Bt (Strength)
2.99 nT
Kp Index
0.67
409
SWS km/s
15:34:00 UTC
-0.3
Bz (nT)
15:35:00 UTC
3.0
Bt (nT)
15:35:00 UTC
Past 24 Hours Data
Past 6 Hours
Solar Wind Speed (km/s)
NOAA / SWPC
14:38
17:39
20:40
23:42
02:43
05:44
08:46
11:47
14:49

Geomagnetic Conditions: 4 AUG 2026

Solar wind:
Bz = -0.26 nT
speed = 408.6 km/sec
density = 1.46 protons/cm^3

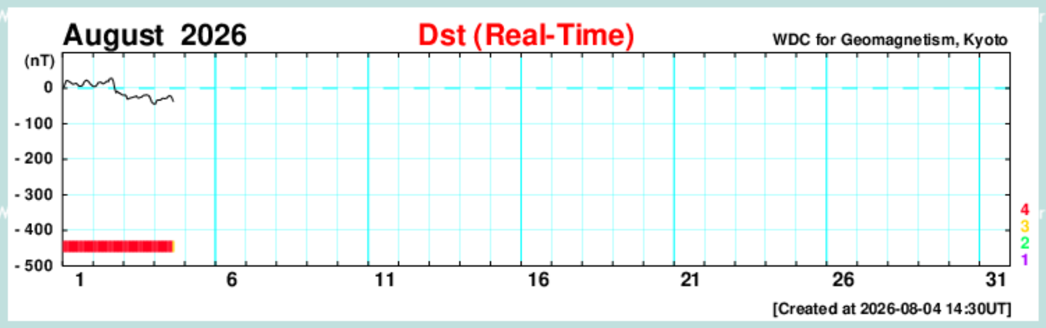
(From – NOAA DSCOVR
In L1, Lagrange Point)



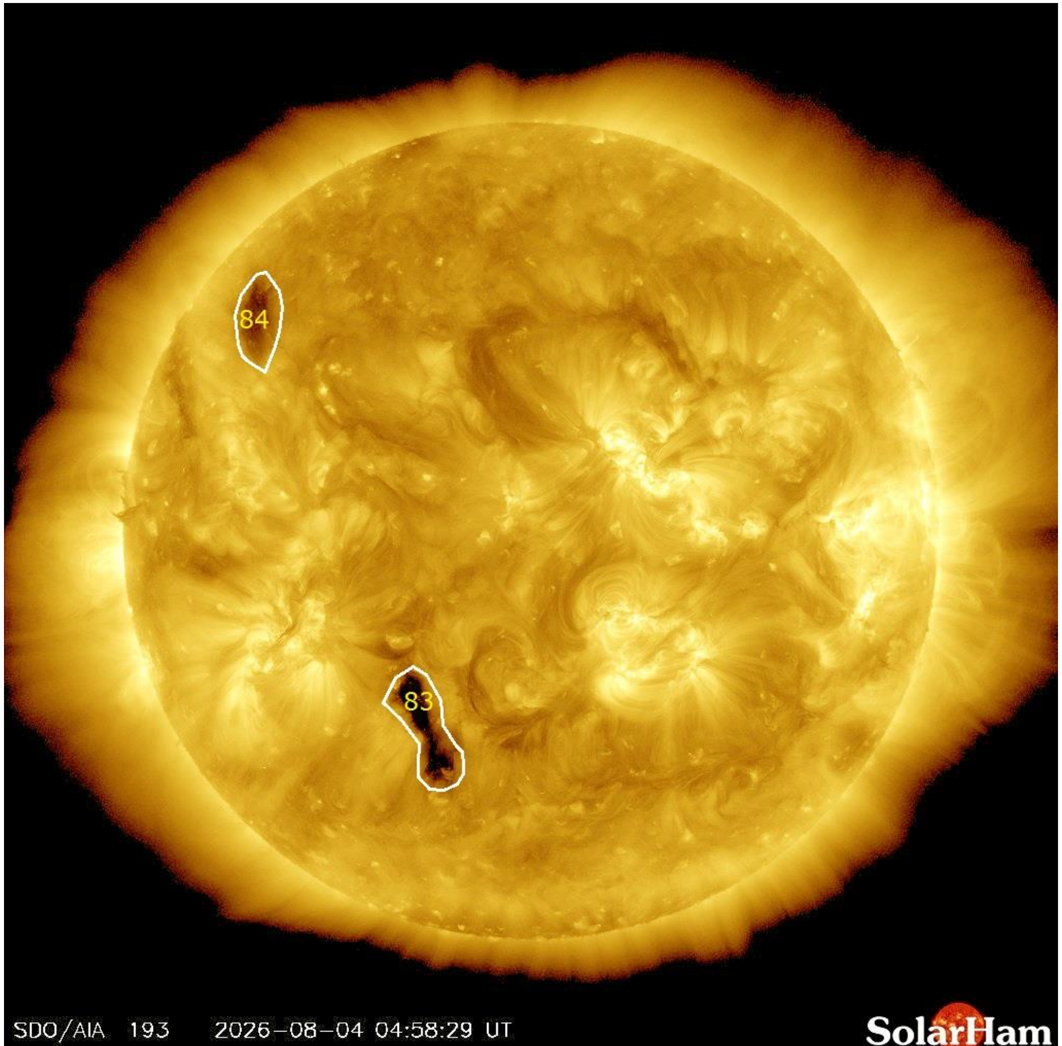
Dst = -28 nT (Ring Field)

(From – Data Analysis Center
For Geomagnetics and Space
Magnetism – Kyoto University)

From – GOES 18, 19
In geostationary orbit



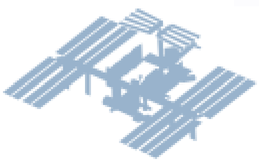
Coronal Holes – 4 AUG 2026

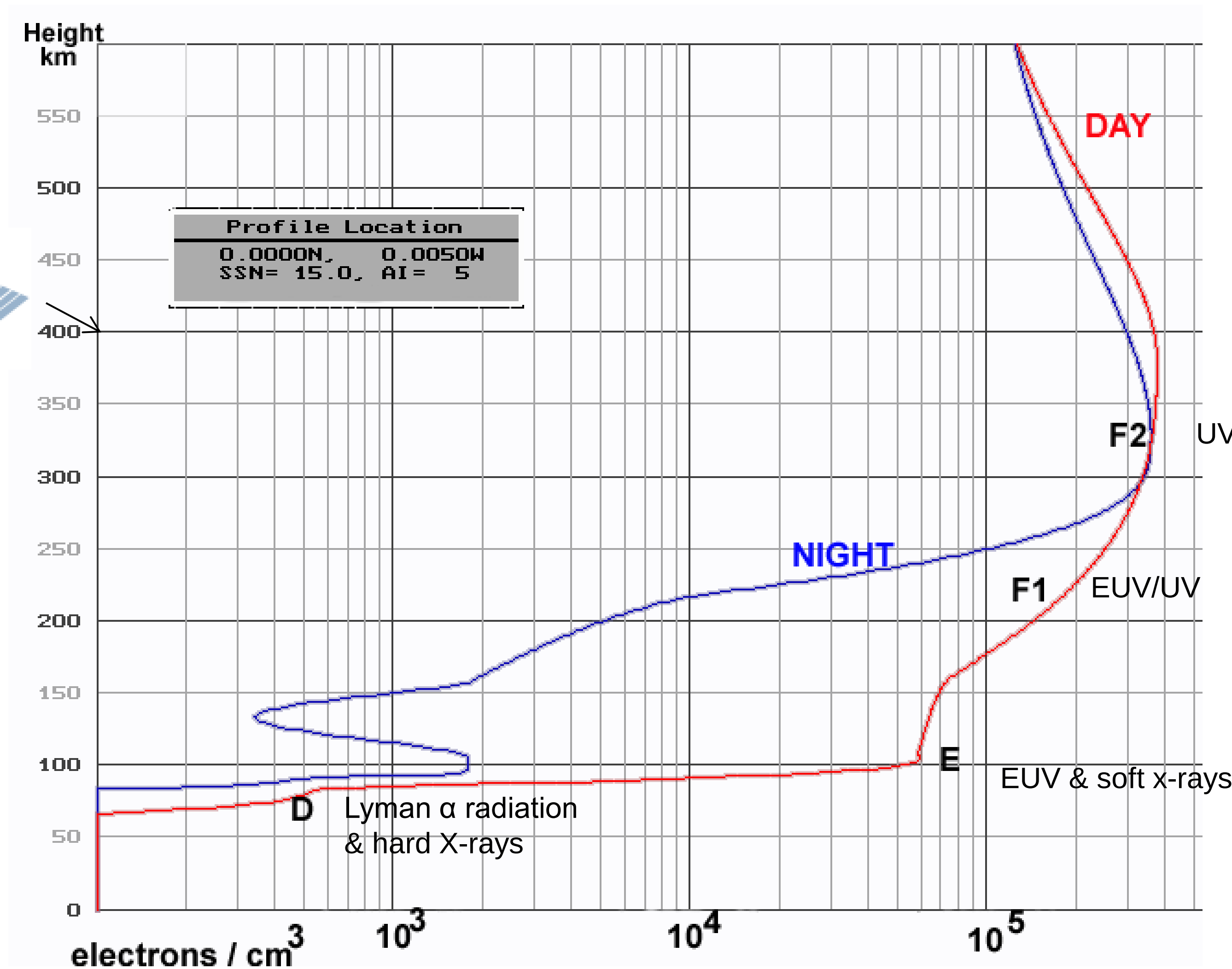


Analysis

There are currently no large coronal holes facing Earth.

Ionosphere Creation





Solar
Radiation

Monoatomic
oxygen

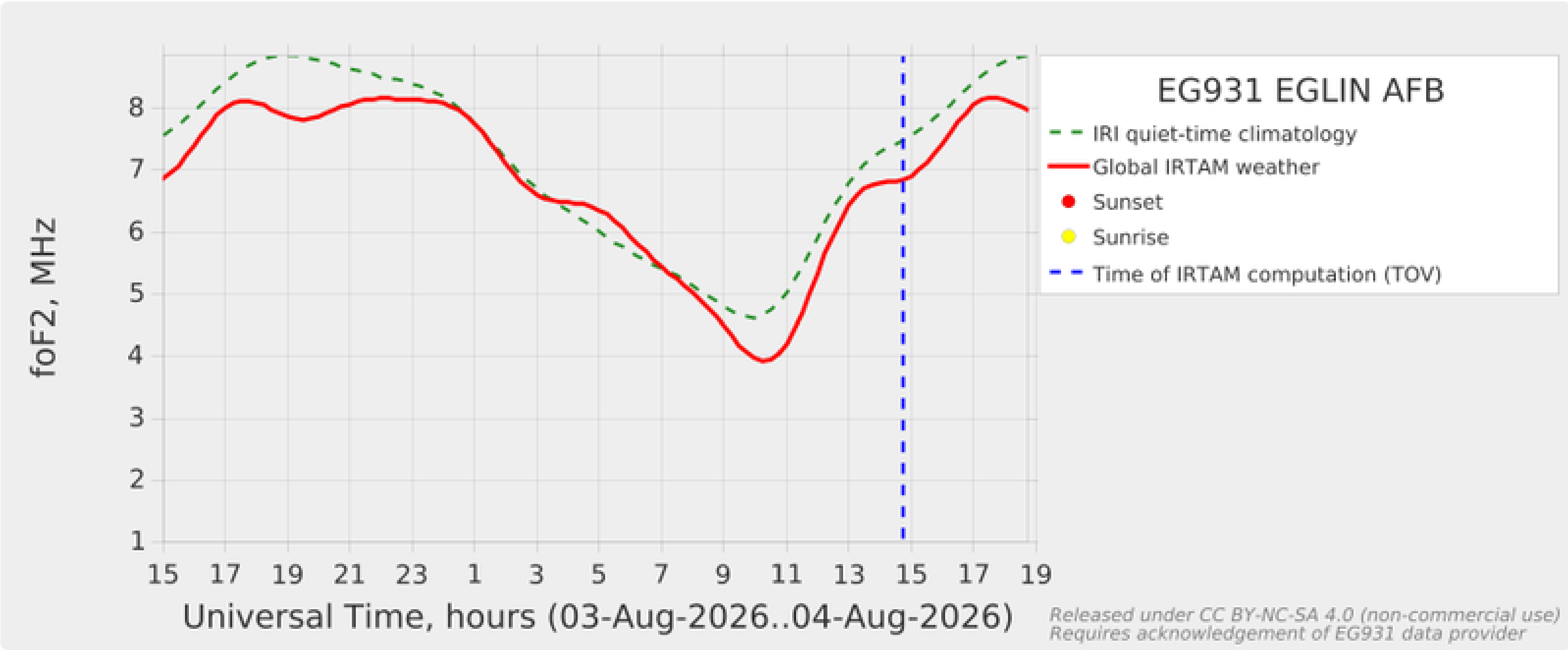
Gravity

Austin Ionosonde Status

- GIRO is presently providing only GAMBIT models for both Austin and Eglin Ionosonde but these foF2 models appear to be accurate, based on MARS net performance (Critical Frequency).

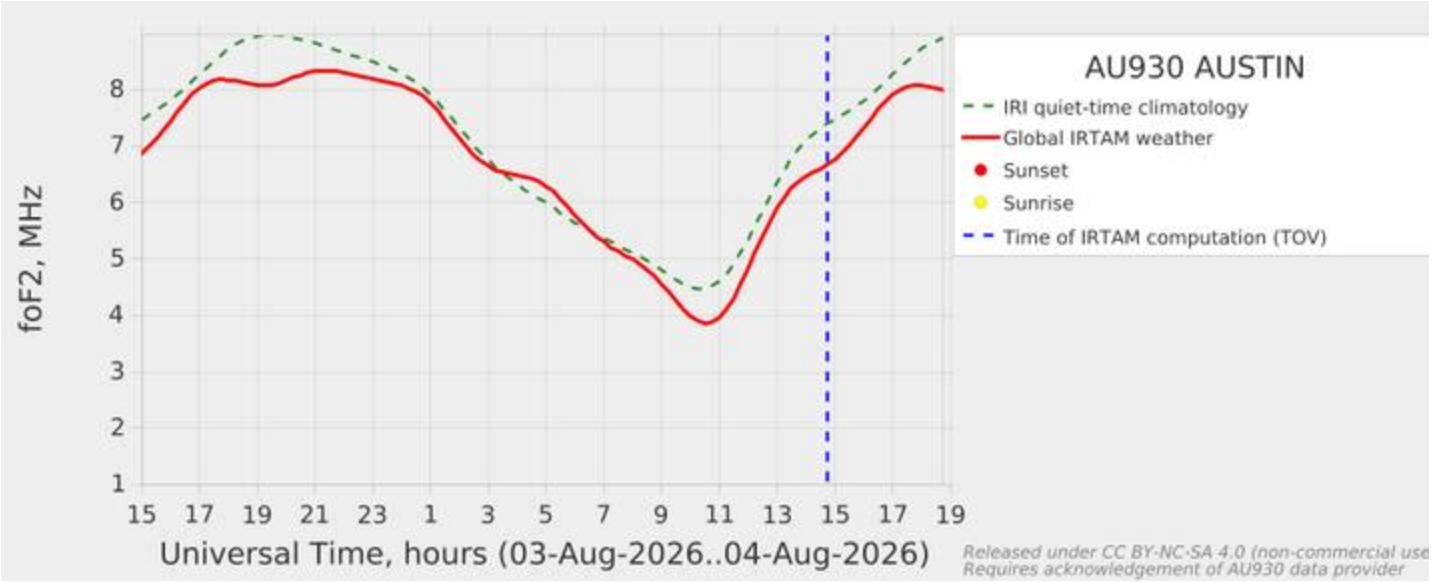
GAMBIT foF2 Trending Chart for Eglin Ionosonde

https://www.region6armymars.org/resources/solarweather.php



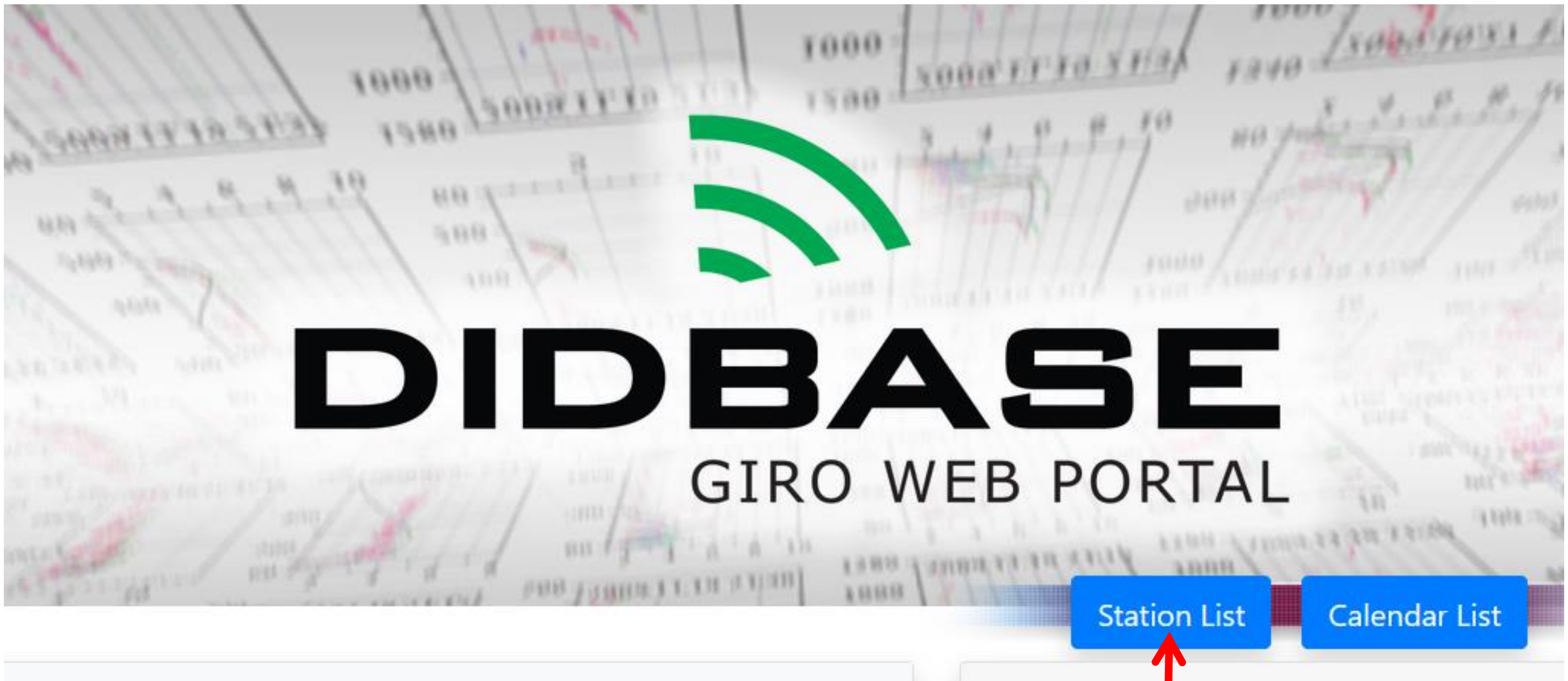
GAMBIT foF2 Trending Chart for Austin Ionosonde – Model only

https://www.region6armymars.org/resources/solarweather.php



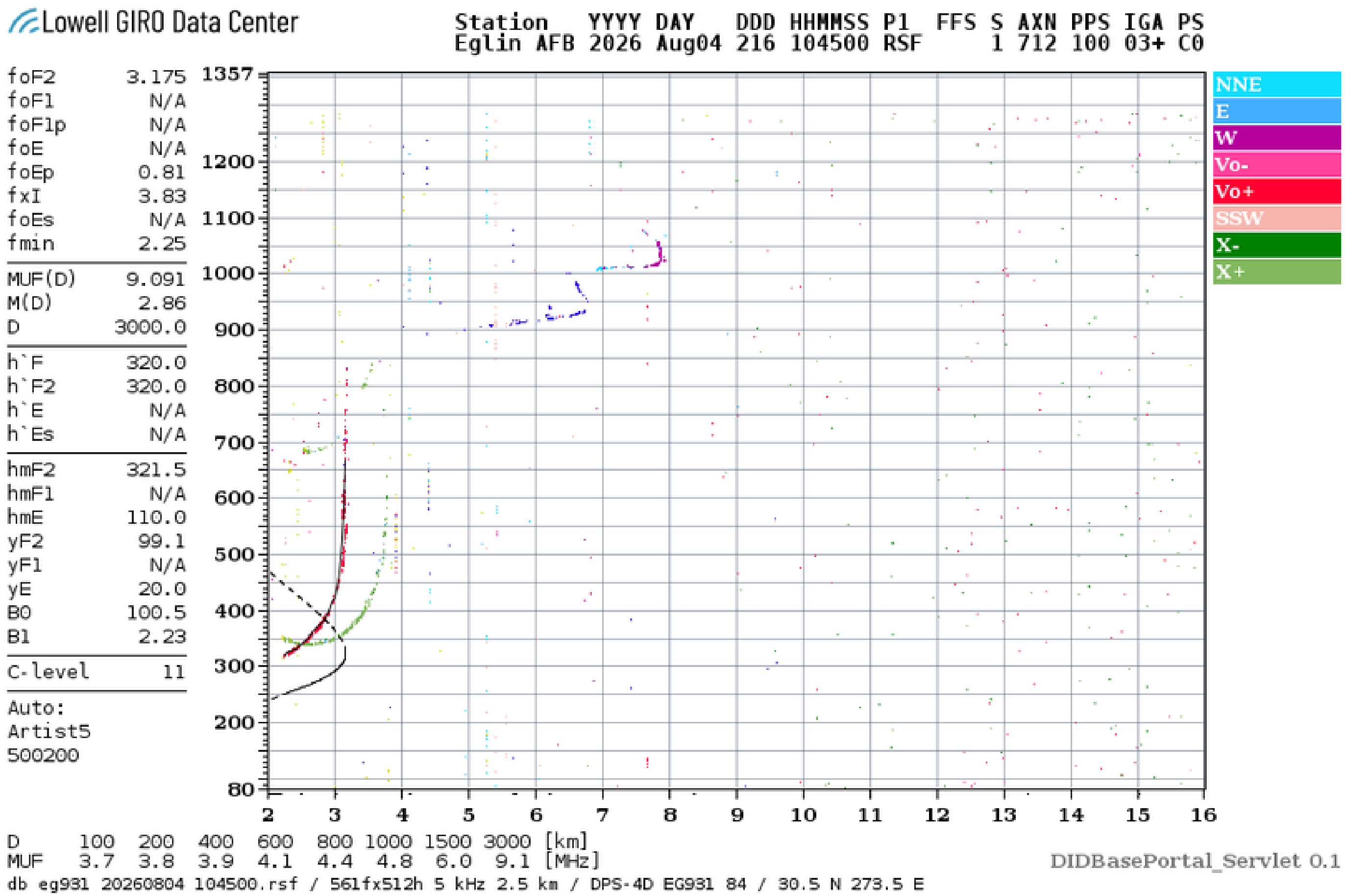
Use of GIRO DIDBASE

https://giro.uml.edu/didbase/



Eglin AFB

Eglin Ionosonde – 10:45 Z



Time Shift for Eglin Ionosonde

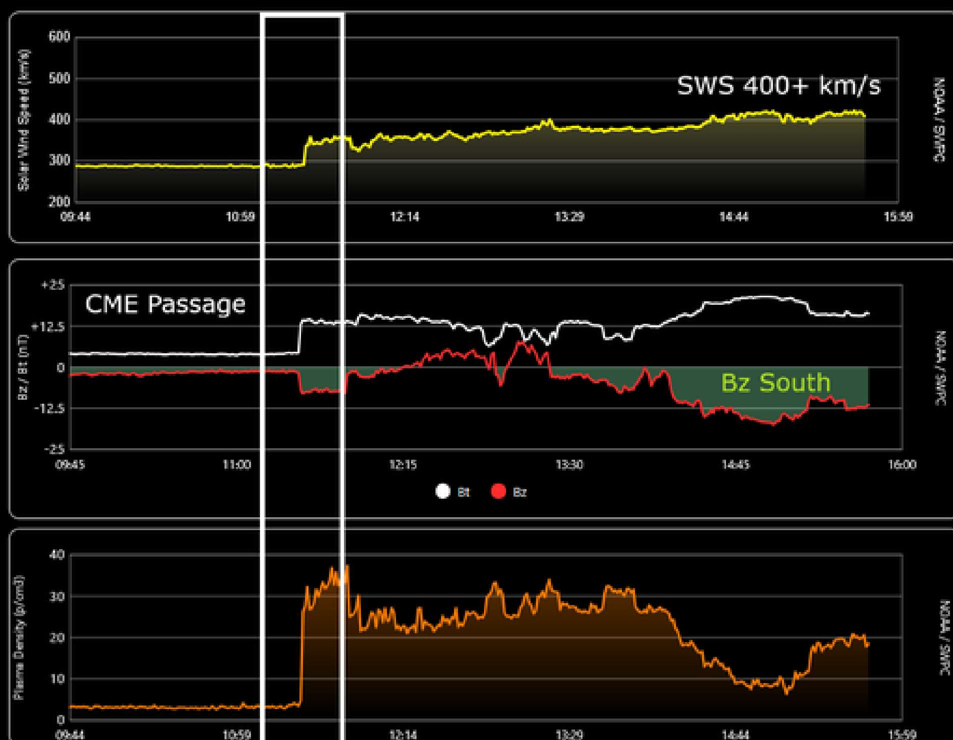
- Eglin AFB Ionosonde is east of Austin at about the same Latitude so sun angle is about the same (same solar insolation).
- What happens to the Ionosphere over Eglin will occur over Austin about 45 minutes later.
- This is simply a Longitude difference converted to time.
- So look at Eglin's foF2 45 minutes earlier to see what is going to happen over Austin.
- The most rapid change in foF2 occurs as the <u>sun is rising and setting (morning and evening).</u>

CME Passage / Moderate Geomagnetic Storm

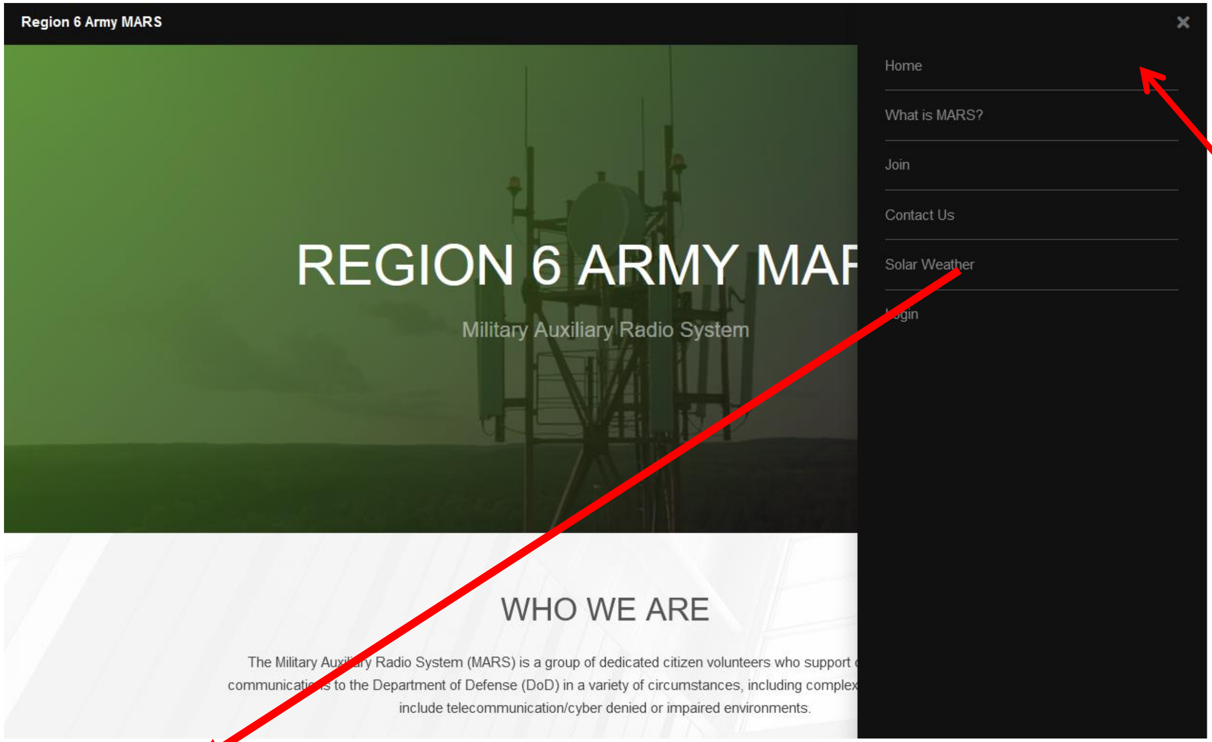
August 2, 2026 @ 17:25 UTC (UPDATED)

Arriving a little later than predicted, the CME associated with the eruption on July 30th reached Earth today at 12:29 UTC (Aug 2). The minor (G1) storm threshold was reached at 17:12 UTC. With the Bz component of the interplanetary magnetic field (IMF) sharply south at times, there will remain a chance for moderate (G2) storming as well. More to follow whenever necessary.

UPDATE: The moderate (G2) storm threshold was reached at 17:59 UTC (Aug 2).



Solar Weather Data



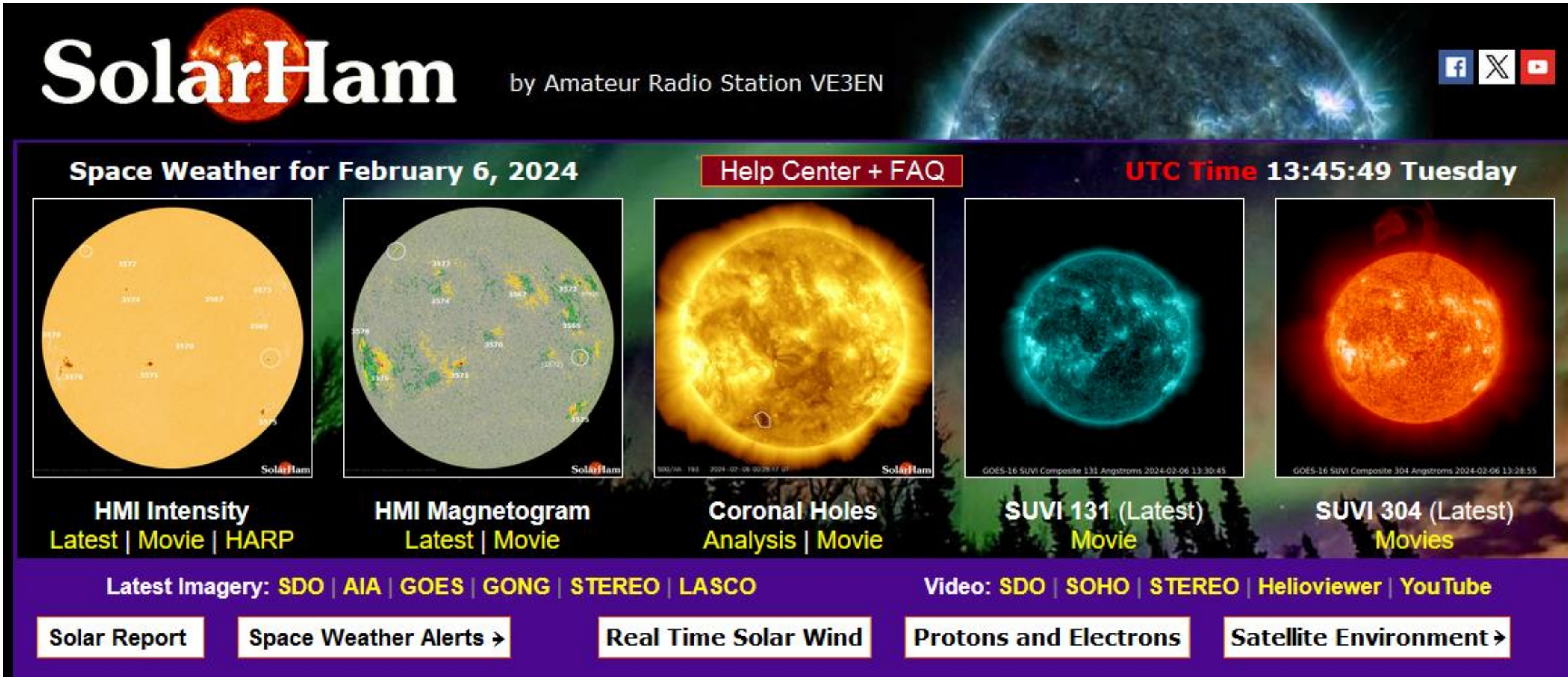
Menu

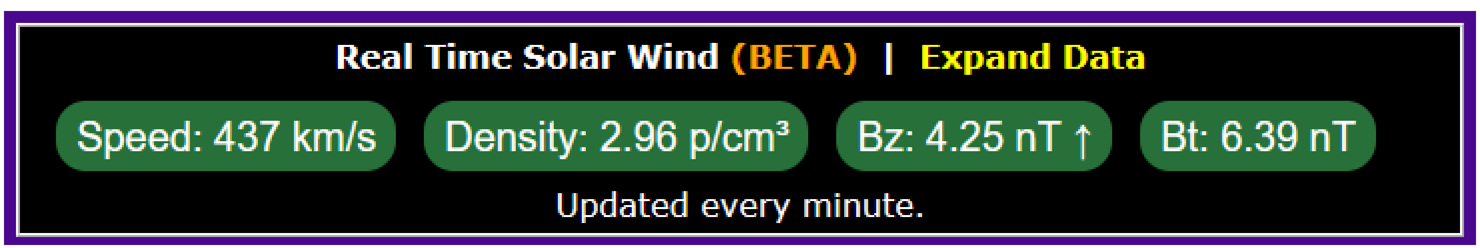
Solar Weather

All Ionosondes
GAMBIT URL

- GAMBIT - Global Assimilative Model of Bottomside Ionosphere Timeline
  - Boulder
  - Eglin

- NOAA Solar Weather - Solar Weather plots of Kp and X-Ray and other solar emissions.
- Solen Solar Weather - Good general solar forecast from an individual.
- Solar Ham - SolarHam provides real time solar news, as well as consolidated data from various sources.


SolarHam
by Amateur Radio Station VE3EN
Space Weather for February 6, 2024
Help Center + FAQ
UTC Time 13:45:49 Tuesday
HMI Intensity
Latest | Movie | HARP
HMI Magnetogram
Latest | Movie
Coronal Holes
Analysis | Movie
SUVI 131 (Latest)
Movie
SUVI 304 (Latest)
Movies
Latest Imagery: SDO | AIA | GOES | GONG | STEREO | LASCO
Video: SDO | SOHO | STEREO | Helioviewer | YouTube
Solar Report
Space Weather Alerts
Real Time Solar Wind
Protons and Electrons
Satellite Environment


Real Time Solar Wind (BETA) | Expand Data
Speed: 437 km/s
Density: 2.96 p/cm³
Bz: 4.25 nT ↑
Bt: 6.39 nT
Updated every minute.

spaceweather.com

News and information about the Sun-Earth environment

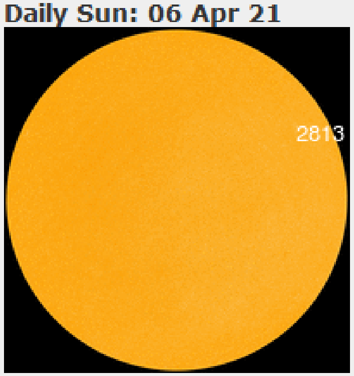
https://www.spaceweather.com/

Current Conditions

Solar wind
speed: **314.8** km/sec
density: **9.9** protons/cm^3
more data: ACE, DSCOVR
Updated: Today at 1225 UT

X-ray Solar Flares
6-hr max: **A1** 1027 UT Apr06
24-hr: **A1** 1515 UT Apr05
explanation | more data
Updated: Today at: 1230 UT

Daily Sun: 06 Apr 21



Sunspot AR2813 is decaying, and poses no threat for strong flares. Credit: SDO/HMI

FLYING TO THE VOLCANO: Iceland's Geldingadalur volcano has turned into an popular tourist attraction---especially since auroras were sighted above the glowing lava. Early this morning, Tuesday, April 6th, Brian Emfinger saw auroras before he even reached the Reykjanes peninsula:

QUESTIONS?

Lewis Thompson

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